University of Massachusetts Boston and Amherst

STEMPOWER FELLOWSHIP

Institute of Diversity Sciences

Funded by the Alfred P. Sloan Foundation

FACULTY BROCHURE

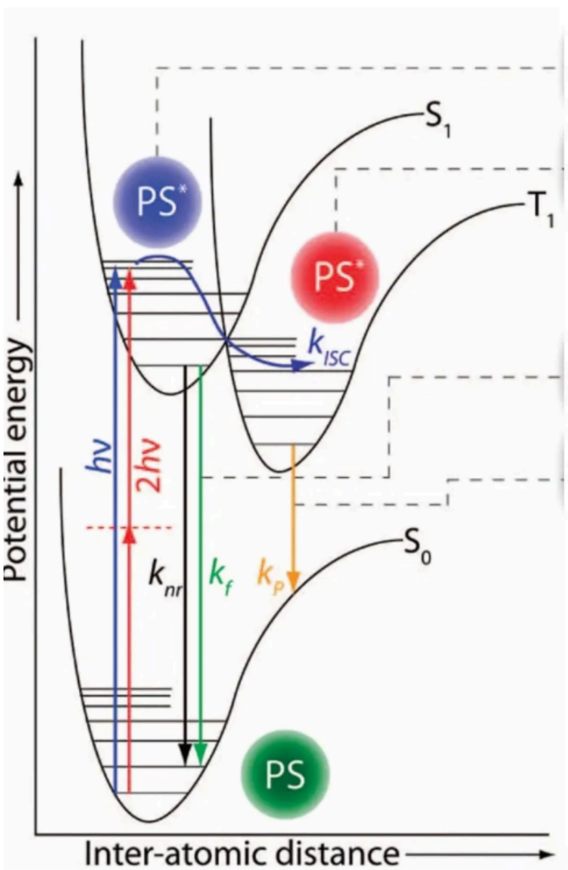
Fall 2026

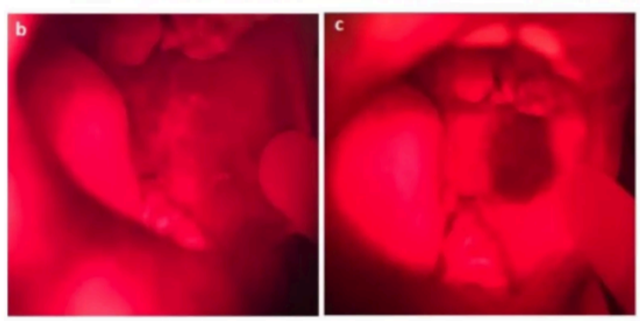
More Information: STEMPOWER.net

Department: Physics

Jonathan Celli

Our laboratory has ongoing projects to develop enabling technology for clinical photodynamic therapy (PDT) as well as basic science studies aimed at elucidating biological and biophysical determinants of PDT efficacy in preclinical models of human cancers. We have made significant contributions to the use of bioengineered 3D tumor models for therapeutic screening and have developed new technologies for tumor imaging and PDT. A major area of focus is on clinical translation of image-guided PDT as a cancer treatment technology for global health settings.



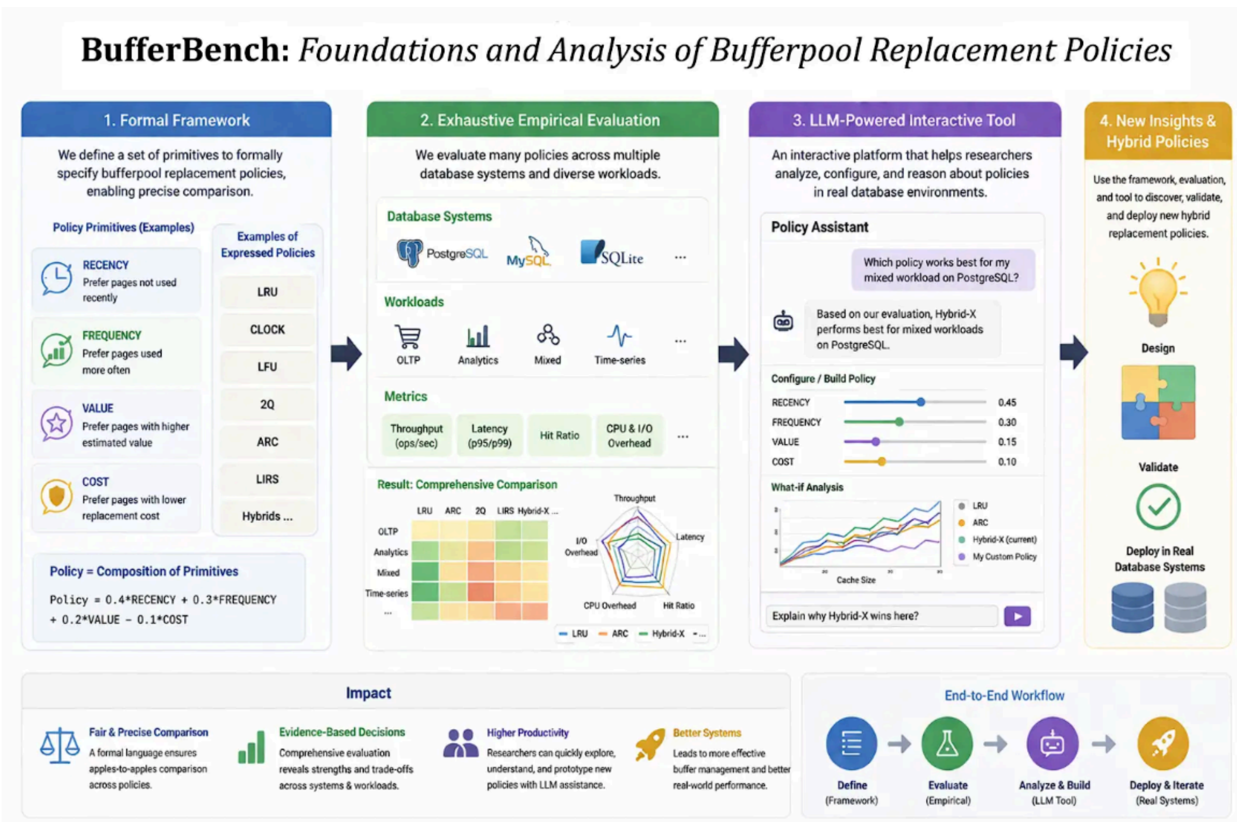


Website: https://www.quantum.umb.edu/Celli/

Department: **Computer Science**

Tarikul Islam Papon

My research focuses on developing novel data management solutions. In this project, we will develop a formal framework for defining database bufferpool replacement policies using a set of primitives, enabling precise comparison across numerous existing policies. Building on this, we will conduct an exhaustive empirical evaluation across multiple database systems, measuring performance under diverse workloads. We will also develop an interactive demonstration tool powered by large language models, helping researchers to analyze, configure, and reason about replacement policies in real database environments and build new hybrid page replacement policies.

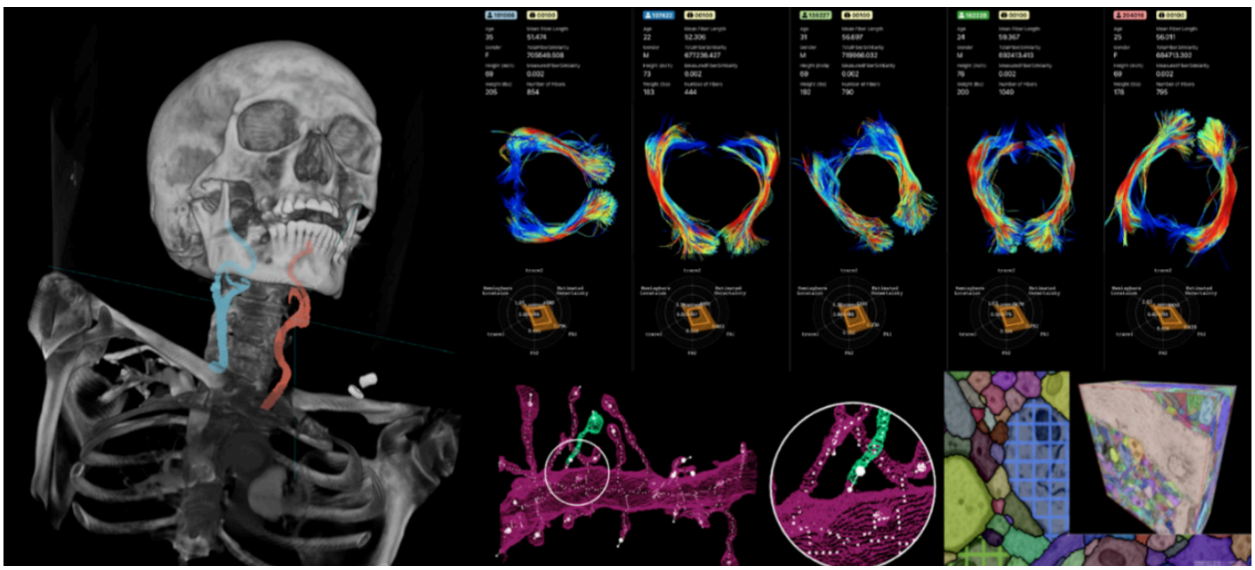


Website: https://www.cs.umb.edu/~tpapon/

Department: **Computer Science**

Daniel Haehn

The Machine Psychology research group at UMass Boston develops visual computing methods to accelerate biological and medical research Our group also manages the Artificial Intelligence Research Core with applications in Data Science, High-Performance Computing, and Cybersecurity.



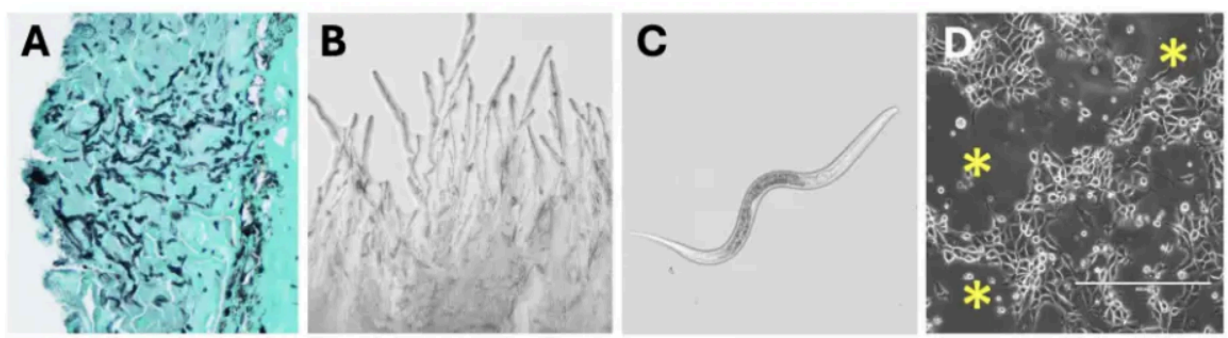
Website: https://danielhaehn.com/

Department: Biology

Douglas Woodhams

Developing rapid diagnostics for an emerging amphibian disease

The discovery of an unknown and highly transmissible salamander disease has triggered an inter-agency emergency response, yet field surveillance and containment remain severely constrained by the absence of a reliable diagnostic tool. The goal of this proposal is to develop and validate a high-throughput quantitative PCR (qPCR) assay to detect the causative agent of this emerging amphibian disease. Using pathogen genomic data generated from ongoing infection trials and metagenomic sequencing, we will design probe-based assays for sensitive, specific, and noninvasive detection from skin swabs and environmental DNA samples.



Website: https://woodhamslab.com/

Department: **Chemistry**

Qingjiang (Vince) Li

Our research interests are at the interface of the chemistry and biology of carbohydrates, including organic synthesis, enzymatic synthesis of complex carbohydrate molecules and their applications in immunology. Through multidisciplinary research, students are vigorously trained with multi-step chemical and enzymatic synthesis and exposed to collaboration in immunology and glycobiology.

Synopsis

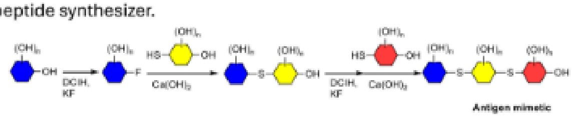
Carbohydrates play important roles in many fundamental biological processes, such as inflammation, microbial infections, tumor metastasis and immune responses. At the center of investigations in these areas is the study of carbohydrate–protein interactions, which is hindered by the inaccessibility of oligosaccharides and glycoconjugates in well-defined form. Our research interests are at the interface of the chemistry and biology of carbohydrates, including organic synthesis, enzymatic synthesis of complex carbohydrate molecules and their applications in immunology. Aiming at changing the feedstock of carbohydrates to related biological studies, the Li lab is developing chemical and chemoenzymatic methods to expedite the assembly of oligosaccharide and mimics. The resulting glycans or mimetics are exploited as component of therapeutic vaccines for various diseases including cancer. There are three research areas currently being explored:

Development of new glycosylation method

Carbohydrate synthesis is featured with tedious protection/deprotection process. We are exploring heterocycles as prospective leaving groups that selectively react with anomeric hydroxyl group to alleviate the labor-intense protection process. The environment-friendly, protection-free fluorination reaction shown below is currently being developed in the laboratory.

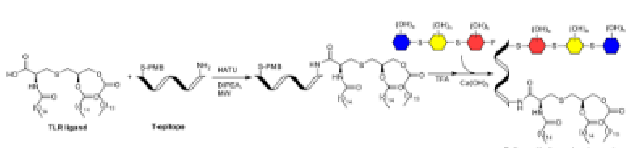
Synthesis of glycomimetics

The major challenge in oligosaccharide synthesis is the stereoselective construction of glycosidic linkage between monosaccharides. Our goal is to find replacement of glycosidic bond to realize the facile assembly of glycomimetics in streamlined manner with the ultimate goal of automation by commercial peptide synthesizer.



Development of carbohydrate-based vaccine

The glycomimetic *S*-glycosidic bond closely resembles Tumor-Associated Carbohydrate Antigens (TACAs), while being more immunogenic, metabolically stable and easy to assemble. They will be conjugated onto carriers for the construction of multi-component vaccine against cancer.

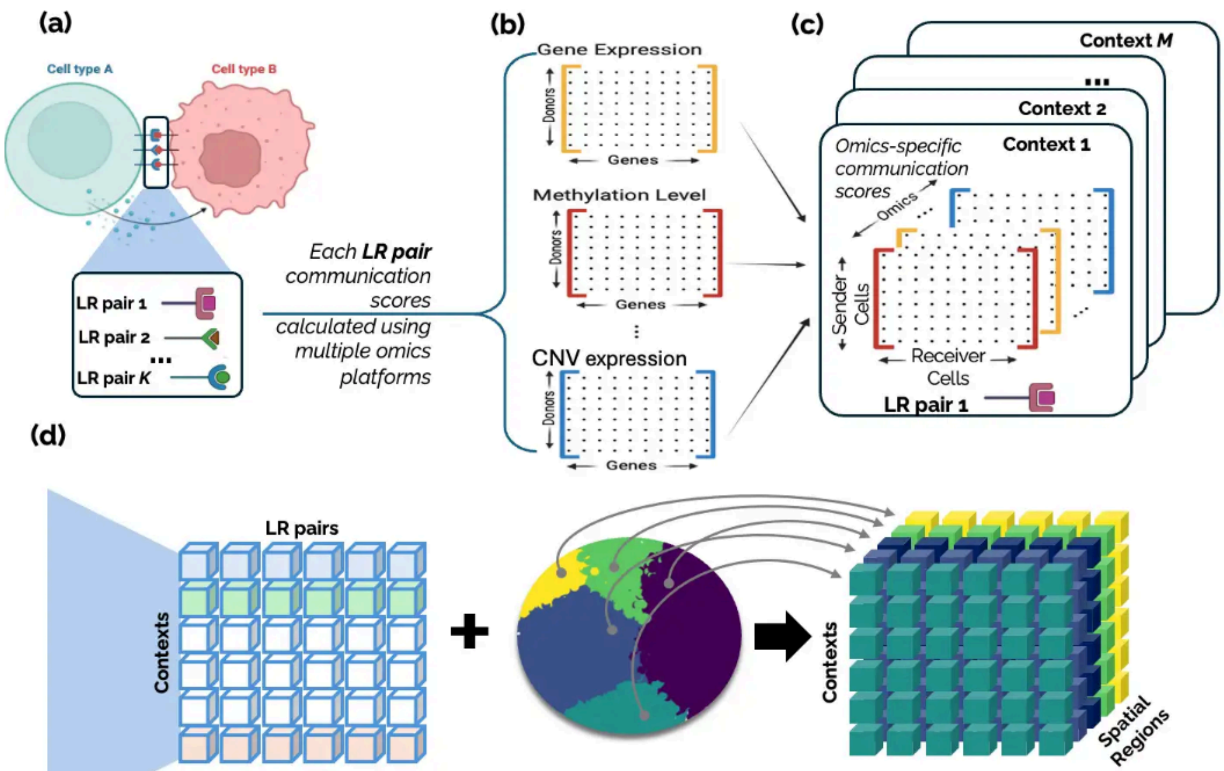


Website: https://alpha.chem.umb.edu/faculty/qingjiang.li/home/

Department: **Mathematics**

Neriman Tokcan

My research develops machine learning methods with a specific focus on tensor-based frameworks for integrating multi-omics and spatial transcriptomics data to study cancer biology and cell–cell communication. I investigate biologically meaningful patterns across heterogeneous biomedical datasets, with applications to tumor microenvironment analysis, cellular interactions, and disease progression. My work combines applied mathematics, data science, and computational biology to develop scalable and interpretable computational tools for high-dimensional biological data and translational biomedical research.



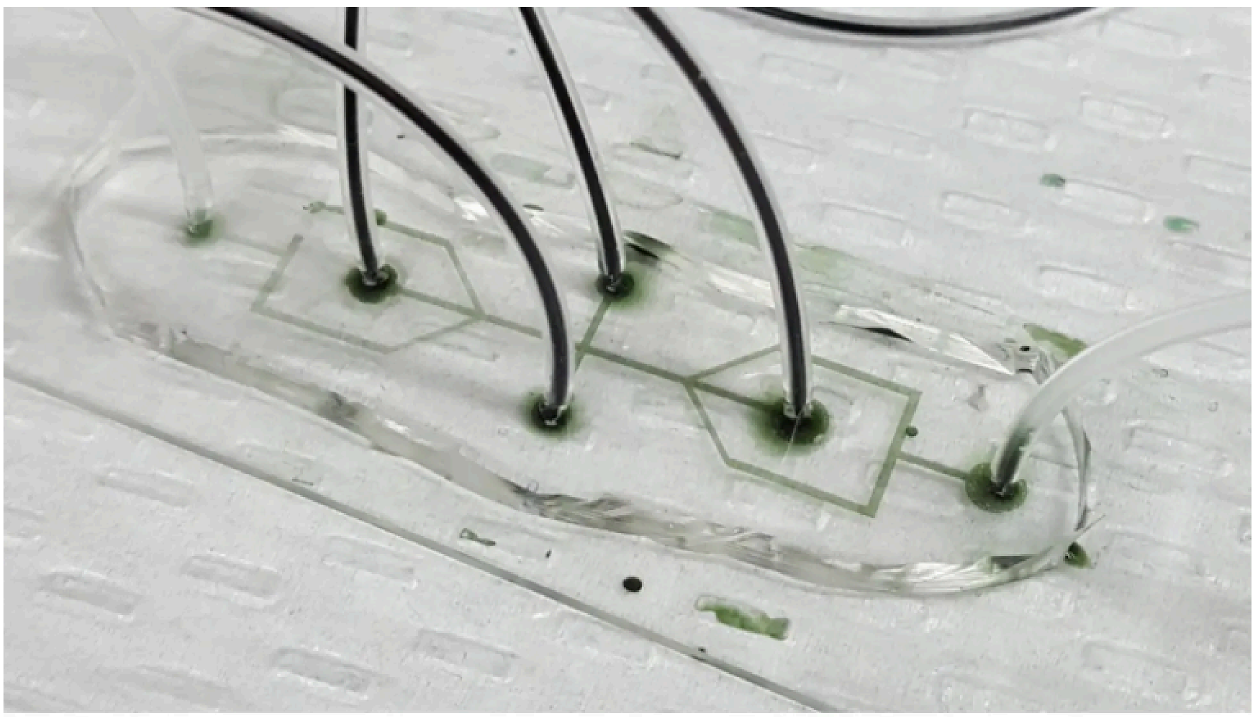
Multimodal tensor construction for modeling cell–cell communication across biological contexts and spatial structure

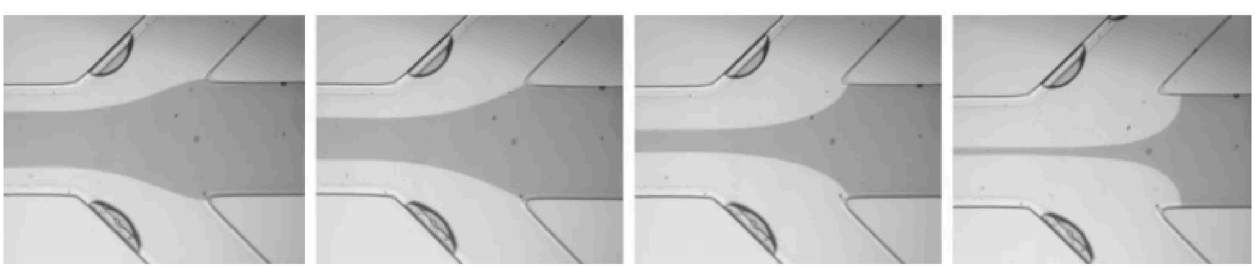
Website: https://nerimantokcan.com/

Department: Engineering

Joanna Dahl

The Biomechanics Lab at UMass Boston (PI Joanna Dahl) contributes fundamental understanding of microscale cell and vesicle biomechanics by measuring their intrinsic stiffnesses. We have developed microfluidic tools that are well suited for these small, soft objects found in biology. We study a promising biomarker, extracellular vesicles, and have begun an investigation of cell clusters to bridge the gap between single cell and multi-cellular tissue biomechanics.





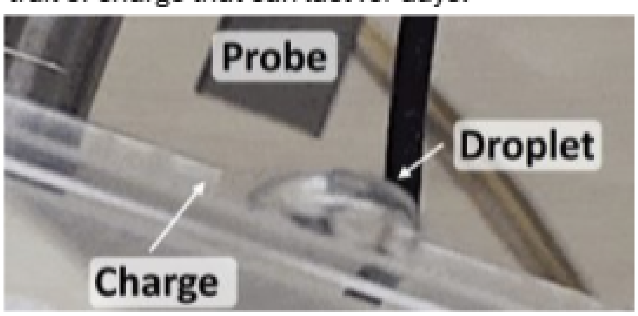
Website: https://sites.google.com/view/umb-biomechanics-lab

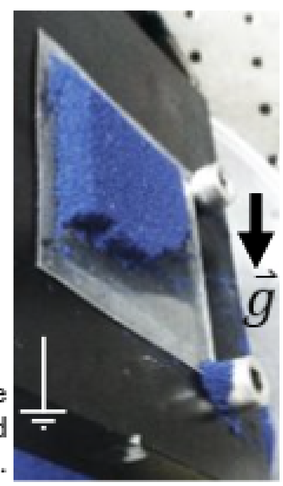
Department: **Physics**

Tony Dinsmore

Our lab studies physical properties of soft matter, such as membranes, fluid interfaces, colloids, emulsions, and powders. Some of our work is inspired by biology: we learn how to design artificial lipid bilayer mebranes that could mimic cellular functions. Other projects are inspired by phenomena that we see all the time around us, yet cannot explain. For example, static charge accumulates on solids and liquids and affects the behavior of powder and other natural materials. Images below show two surprising examples, both of which are relevant to nature and technology.

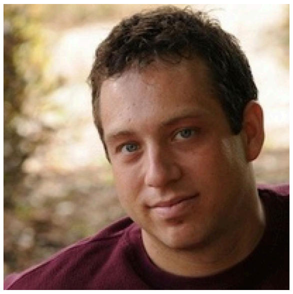
A water droplet slides down a surface, leaving a trail of charge that can last for days.





A pile of like-charged grains can form a rigid, cohesive solid. This mechanism might be useful on the Moon and Mars, which are dry, dusty, and charged.

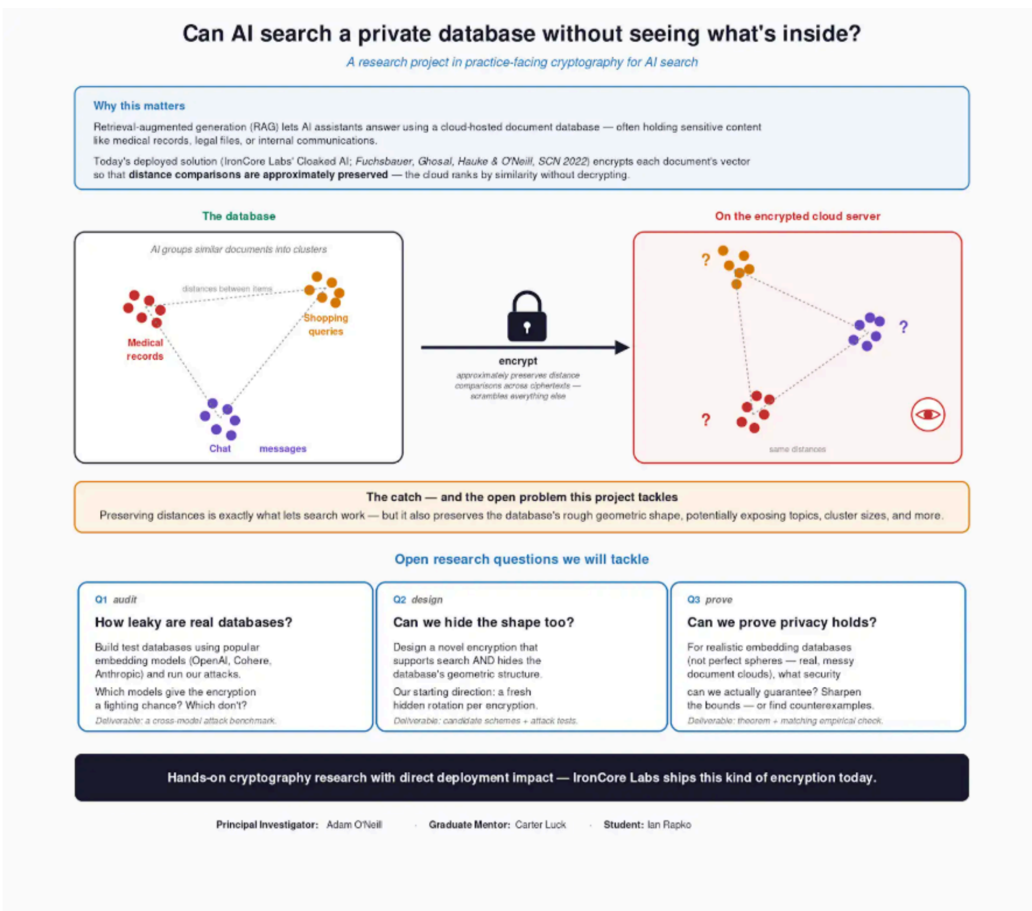
Website: https://websites.umass.edu/dinsmore/



Department: **Computer Science**

Adam O'Neill

Retrieval-augmented generation (RAG) lets AI assistants answer using a cloud-hosted document database---often holding sensitive content like medical records, legal files, or internal communications. Today's deployed solution (IronCore Labs' Cloaked AI; Fuchsbauer, Ghosal, Hauke & O'Neill, SCN 2022) encrypts each document's vector so that distance comparisons are approximately preserved---the cloud ranks by similarity without decrypting. But that also preserves the database's rough geometric shape, potentially exposing topics, cluster sizes, etc. This project will (1) audit how leaky encrypted databases built from popular embedding models are, (2) design stronger encryption for this setting, and (3) prove security for realistic distributions.

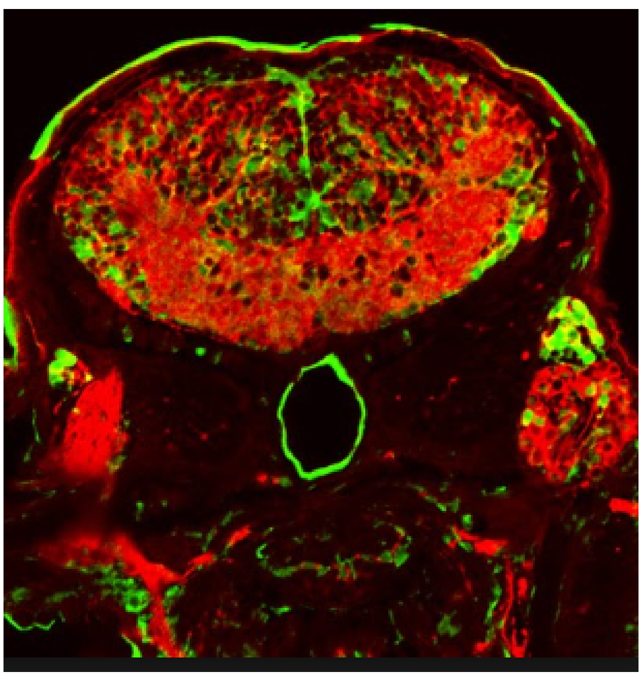


Website: https://groups.cs.umass.edu/oneill/

Department: **Biology**

Gerald Brian Downes

The main goal of our current research is to provide new insights into rare epilepsy disorders and help develop new therapeutics to treat them. To pursue this goal, we use zebrafish, whose developing embryos and larvae offer several features that make them an excellent model system. They develop quickly, exhibit robust motor behavior, and are transparent, allowing their brains to be easily observed. We can also leverage the power of zebrafish genetics to investigate how genes regulate nervous system function. Together, these features enable an integrated approach to studying brain development that combines genetics, microscopic imaging, and behavioral analysis. Because many genes and aspects of brain function are conserved among vertebrates, developing zebrafish offer significant advantages for modeling genetic epilepsies and establishing a high-throughput platform to identify new therapeutic compounds.

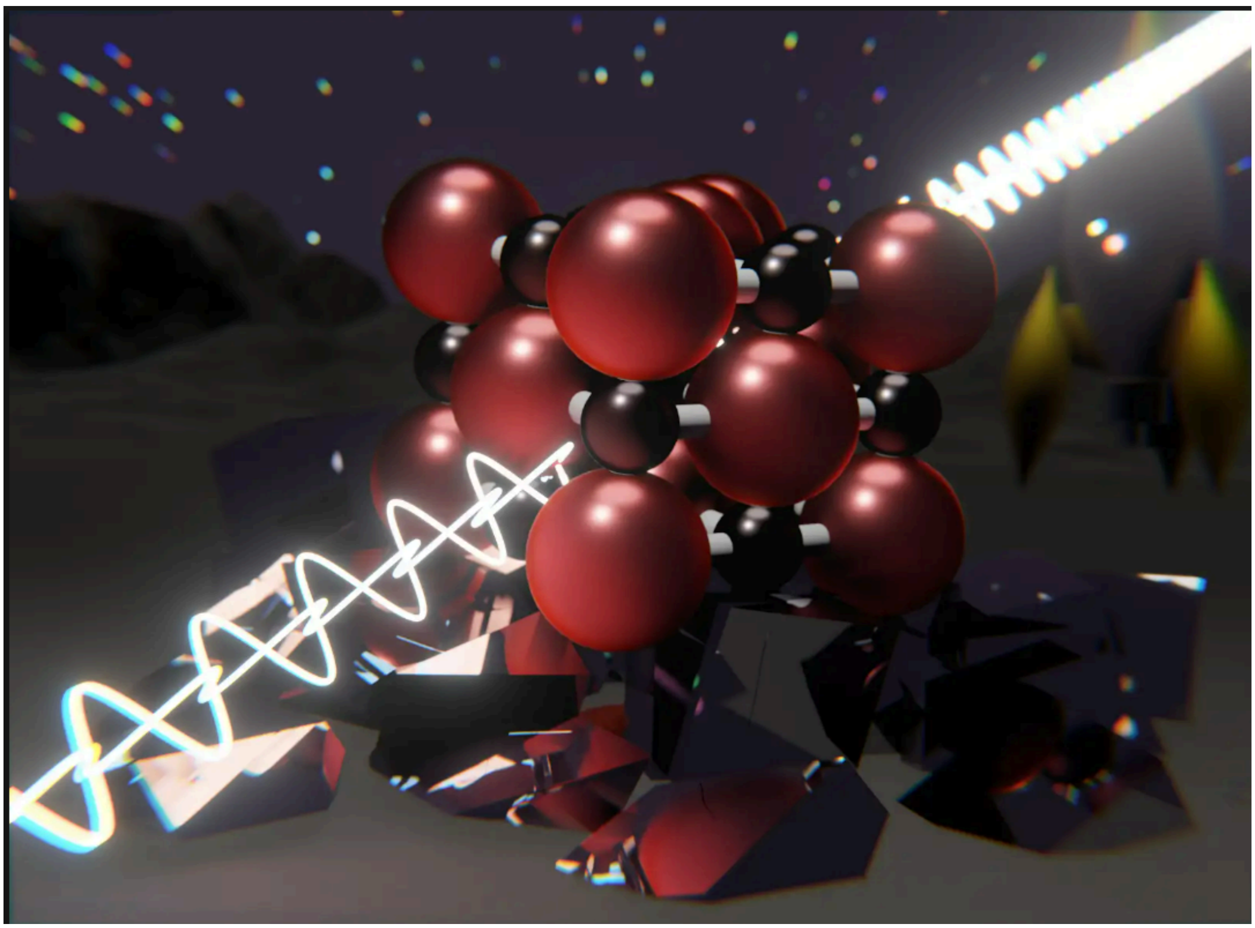


Website: https://www.downeslab.org/

Department: **Chemistry**

James Walsh

We study matter under extreme compression, where conventional rules of chemistry begin to break down and new phases emerge.

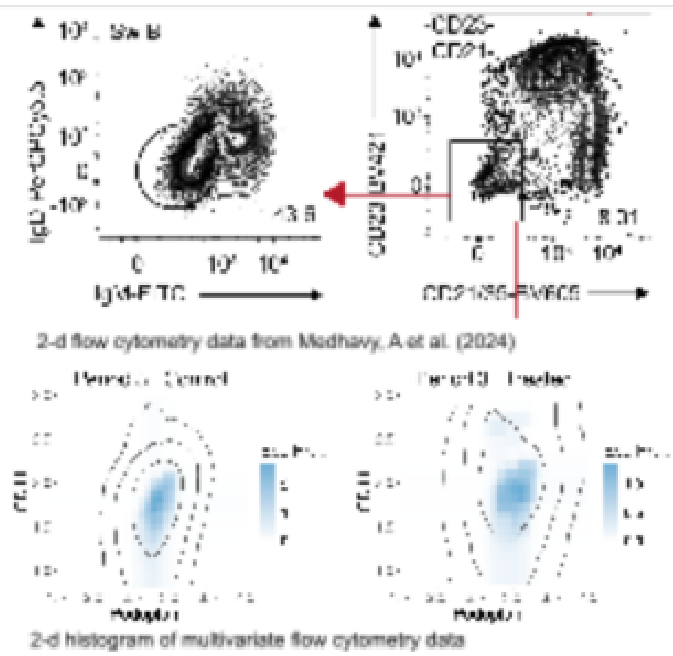


Website: https://walsh.chem.umass.edu/

Department: **Mathematics**

Qian Zhao

Flow cytometry measures biomarkers of tens of thousands of cells simultaneously, allowing researchers to study questions like "how does cell characteristics change as disease progresses". To pinpoint cell populations, scientists often need to manually segment regions using bivariate dot plots, which is time consuming and subjective. An automatic approach can be developed using multivariate histograms. As more biomarkers are measured, traditional histograms with equal sized bins are no longer feasible, and a solution is data-adaptive histogram. In this project, you will experiment with multiple ways to build data-adaptive histograms, and use them to identify cell populations from flow cytometry data.



2-d flow cytometry data from Medhavy, A et al. (2024)

2-d histogram of multivariate flow cytometry data

Website: https://qianzhao14.netlify.app/

Department: **Engineering**

Kara Peterman

We are gluing steel! The research project will be focused on validating a range of structural steels for use in adhesive steel-to-steel construction. Common structural steel grades used across the steel construction industry and high-strength adhesives will be assessed experimentally, complimenting a large-scale experimental effort on steel-to-steel adhesive connections.

Website: https://karapeterman.com/